

TIMOTHÉE CROUZET

Final-Year Engineering Student — Data Science & AI

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Seeking an ML research internship in Sydney · March–August 2027

Eligible for subclass 417 Working Holiday visa · no sponsorship required

SUMMARY

Final-year engineering student specializing in Data Science & AI, with strong training in mathematics and physics and production experience building LLM-based systems at Dust, a French AI startup. Seeking a first academic research experience in machine learning.

SELECTED EXPERIENCE & PROJECTS

Independent Research Project — Oval vs. Round Bicycle Chainrings

Paris · 2023-2024

- Investigated whether oval bicycle chainrings could provide a mechanical advantage; built a Python model of pedaling force and chainring geometry to compare torque profiles.
- Designed experimental measurements to complement the model; found only marginal performance differences under the tested conditions.

Toxic Gas Identification — ML Challenge

ESIEE Paris · 2026

- Built an end-to-end multi-output regression pipeline using pandas, scikit-learn and XGBoost to estimate toxic gas concentrations from sensor measurements.
- Engineered sensor and humidity-interaction features, designed validation splits for distribution shift, and compared regularized models and simple ensembles.

Software Engineer Intern — Dust

Paris · May–August 2026

- 17+ merged PRs to [Dust's open-source production monorepo](#), working on systems for LLM-based agents.
- Built and shipped MCP integrations for Exa, Shopify and Workday, and worked on secure code execution for AI agents in shared sandboxes.

SmartRoute — MTB Route Planner

ESIEE Paris · 2025 Year-End Engineering Project

- Led a five-student team building an MTB route planner using Strava/OpenStreetMap data, a popularity-scoring pipeline and weighted-Dijkstra routing.

EDUCATION

ESIEE Paris — Gustave Eiffel University

2024–2027 (expected)

Diplôme d'Ingénieur in Data Science & AI — Master's-level engineering degree.

Selected Coursework: Fundamental Theory of ML · Machine Learning I & II · Deep & Reinforcement Learning

Lycée Janson de Sailly

Paris · 2022–2024

Selective two-year program with a specialization in mathematics and physics, preparing for competitive entrance exams to top French engineering schools.

TECHNICAL SKILLS

Python & ML — Python, NumPy, pandas, scikit-learn, XGBoost, Matplotlib, feature engineering, model validation.

Software & AI Systems — TypeScript/Node.js, React/Next.js, REST APIs, OAuth, Git/GitHub, PostgreSQL, Docker, MCP, RAG.

LANGUAGES

English — Fluent (TOEIC 955/990, CEFR C1) · French — Native · Spanish — Working knowledge

INTERESTS

Tennis (team captain, interclub) · Football (organised tournaments) · Skiing (raced ages 12–16)